Remote Sensing-Oriented Small-Object Detection Based on Rotation-Normalized Prompting Segment Anything Model

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***Abstract*—Remote sensing object detection is critical for Earth observation and analysis. However, accurately detecting small objects remains a significant challenge due to complex and heterogeneous backgrounds as well as the inherently limited information these objects provide. While numerous specialized algorithms have been proposed to enhance small-object detection, the weak feature representation of objects within cluttered scenes continues to severely limit performance, making this a persistent and open research problem. Therefore, in this article, to enhance the discriminative representation of small objects in a complex background, a novel two-phase rotation-normalized prompting-oriented small-object detection (RNP-OSOD) framework is proposed. In the first mask generation phase, a rotation-normalized prompt for segment anything model (RNP-SAM) is proposed to obtain high-quality instance and semantic mask annotations. In this method, image patches containing objects are cropped, enlarged, and rotation-normalized to acquire RNP before being fed into SAM, which then produces accurate mask supervision for training. In the second mask supervising object detection phase, to amplify the feature responses of small objects and balance instances across different scales, a scale-adaptive instance focusing (SAIF) loss is proposed by employing the generated masks as supervisory signals. This loss comprises an area-guided semantic reconstruction loss, called scale-adaptive semantic (SAS) loss and a detection-decoupled instance segmentation loss. To rigorously evaluate the proposed framework, we conduct extensive experiments on the small object detection datasets (SODAs) and create a new dataset, Small-DOTA, constructed by filtering and downsampling DOTA to match SODA's characteristics. The experimental results demonstrate that our method achieves state-of-the-art performance across both benchmarks. The source code will be available at https://github.com/cbq233333/RNP-OSOD**

***Index Terms*—Auxiliary supervision, remote sensing small-object detection, segment anything.**

I. Introduction

REMOTE sensing object detection aims to locate and identify potential objects of interest in aerial or satellite imagery, serving as a foundational task in the interpretation of remote sensing images. It has catalyzed a wide range of critical applications, including security monitoring [1], traffic supervision [2], urban planning [3], and precision agriculture [4], underscoring its significant academic and practical value.

The advent of deep learning has dramatically propelled the development of object detection techniques. These methods are generally categorized into two-stage and one-stage frameworks, distinguished by whether they extract region proposals. Two-stage detectors first generate region proposals and then classify them. They typically achieve higher accuracy, especially for small objects [7], but at the cost of slower inference speeds [8]. In contrast, one-stage detectors perform localization and classification simultaneously, prioritizing computational efficiency, although often at the expense of reduced accuracy compared to their two-stage counterparts [8].

Despite these advancements, small-object detection remains a formidable challenge in remote sensing [9]. Compared to medium and large objects, small objects often comprise only a few pixels. They suffer from a coarse appearance, a lack of discernible internal features, and low contrast against complex backgrounds. Furthermore, their frequent dense distribution leads to severe occlusion, making it exceedingly difficult for detectors to distinguish and separate individual instances. Due to these inherent characteristics, existing one-stage and two-stage detection frameworks struggle to learn effective features, leaving them vulnerable to background noise.

In response to these challenges, the mainstream approach has been to extend existing detection networks by designing feature enhancement modules and sampling strategies tailored for small objects. These designs include attention mechanism [10], [11], contextual information exploitation [12], [13], [14], and reselecting high-quality positive samples for small object [5], [15]. However, as shown in Fig. 1(a), the complex backgrounds of remote sensing images and the blurred textures of small objects still result in a substantial number of missed detections, even for the state-of-the-art method, that is, CFINet [5].

Intuitively, incorporating object mask information that defines an object's precise shape and boundary could provide powerful supervisory signals, guiding the detector to enhance the model's learning capability for discriminative representation of small objects against complex remote sensing backgrounds. However, acquiring high-quality pixel-level mask annotations for objects in remote sensing images is prohibitively expensive and labor-intensive, creating a major

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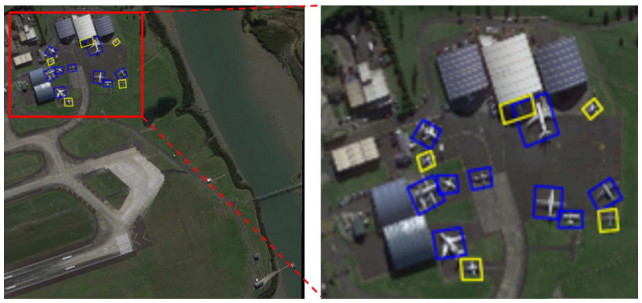
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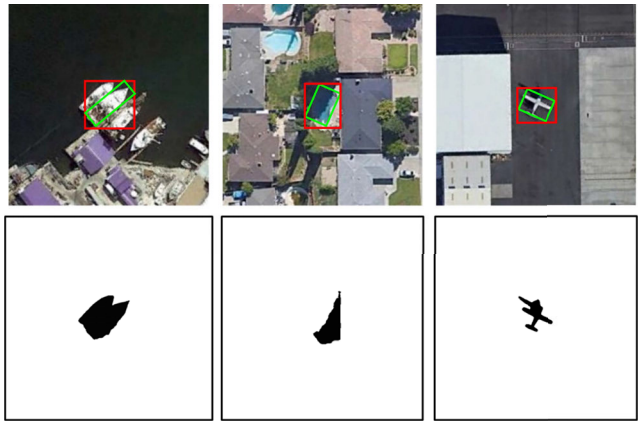
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(a) Limitation of state-of-the-art small object detection method, CFINet [5] on Small-DOTA dataset.



(b) Limitation of state-of-the-art object mask generation method with minimum circumscribed horizontal box prompts proposed by Wang et al. [6] on SODA-A dataset.

Fig. 1. (a) Detection results of the state-of-the-art method CFINet on Small-DOTA dataset. The blue box represents true detection (TP), and the yellow box represents missing detection [true negative (TN)]. It can be seen that, in a complex background, this method produces a significant number of missed detections. (b) Segmentation results of SAM with minimum circumscribed horizontal box prompts on the SODA-A dataset. The green box represents rotated bounding box of the object, and the red box represents the minimum circumscribed horizontal bounding box of it. The minimum circumscribed horizontal bounding box introduces redundant background information, e.g., other densely arranged objects, object occlusions, and object shadows.

bottleneck for this solution. Recently, the segment anything model (SAM) [16] presents a groundbreaking opportunity to circumvent this issue. Pretrained on enormous and diverse datasets, SAM exhibits exceptional zero-shot generalization capabilities for generating pixel-level masks. Therefore, for small objects within the complex remote sensing background depicted in Fig. 1(a), providing bounding box annotations to SAM can generate corresponding segmentation annotations offering additional supervisory signals to the detection network. Since SAM does not support a rotated bounding box as prompt, Wang et al. [6] designed the minimum circumscribed horizontal rectangle of a rotated bounding box as a prompt for the rotated object detection dataset. However, as shown in Fig. 1(b), this naive adaptation introduces excessive redundant background information (e.g., adjacent dense objects, occlusions, and shadows), leading to noisy pseudo-masks that degrade rather than enhance detection performance. This highlights a critical unresolved gap: existing SAM prompt engineering for remote sensing fails to align with the geometric characteristics of rotated small objects, resulting in low-quality pseudo-supervision that cannot effectively mitigate small-object detection challenges.

To overcome the aforementioned adverse effects, we introduce the rotation-normalized prompt for SAM (RNP-SAM), a method specifically engineered to optimize pseudo-label generation rather than relying on standard prompting mechanisms. By performing explicit rotational correction and local amplification on object crops, RNP-SAM transcends the limitations of the minimum circumscribed horizontal rectangle used in conventional approaches. This targeted refinement fully leverages the geometric correspondence between the box and the object, providing SAM with a precision-oriented prompt that minimizes background interference. Furthermore, addressing the severe scale imbalance of small objects requires more than generic supervision, which often leads to biased learning. Consequently, we propose the scale-adaptive instance focusing (SAIF) loss, a specialized objective function designed to effectively leverage RNP-SAM's pseudo-labels. The SAIF loss is not a standard application but a composite of two tailored components: a scale-adaptive semantic (SAS) loss, which dynamically adjusts global feature weights based on instance area to prioritize small objects, and a detection-decoupled instance segmentation loss that sharpens the model's sensitivity to structural details like shape and boundaries. These distinct, problem-specific improvements culminate in the rotation-normalized prompting-oriented small-object detection (RNP-OSOD) framework—instantiated via the classic region of interest (RoI) Transformer—to significantly enhance the discriminative representation of small objects in complex remote sensing scenes. Unlike straightforward adaptations, this work integrates geometric reasoning into prompt design and unifies global semantic balancing with local instance detail supervision, creating a synergistic system that addresses the unique challenges of remote sensing small-object detection. The main contributions of this work are summarized as follows.

1) The RNP-OSOD framework includes two phases, i.e., mask generation and mask supervising object detection, and is proposed to use high-quality pseudo-segmentation labels to provide dense supervision, guiding the detection network to highlight both the small object and its details under a complex background. To achieve this, in the first mask generation stage, the proposed RNP-SAM aims to generate pixelwise pseudo-segmentation annotations for the object detection task. In the second mask supervising object detection stage, the pseudo-labels are employed as supervisory signals at both the global feature level and the local feature level for SAIF loss. In this way, the small-object detection ability of a detection network can be improved.
2) The RNP-SAM is proposed to obtain high-quality pseudo-segmentation labels by minimizing background interference in the box prompt. In the RNP-SAM, the object slice is cropped based on the object center and the predefined context range. The enlarged object slice and rotated bounding box are then normalized to a horizontal state according to the angle of the box to obtain the rotation RNP. By inputting the normalized object slice

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and RNP, SAM can generate high-quality object masks as pseudo-segmentation labels for supervising network training.

3) The SAIF loss is proposed to utilize pseudo-segmentation labels generated by RNP-SAM to assist in supervising the training process and learn fine-grained feature representations of small objects at different scales. In SAIF loss, SAS loss measures the discrepancy between the semantic segmentation results predicted by global features and pseudo-labels. It applies higher loss weights to smaller instances based on the sigmoid function, thereby directing the model to prioritize smaller objects during convergence while balancing the quantitative disparity between objects of varying scales. And the instance segmentation head decoupled with the detection head is incorporated into the detection network, in which the instance segmentation loss measures the difference between the instance segmentation results predicted by local features and the pseudo-labels, strengthening the model to focus on small-object details such as object shape and boundaries.

Extensive experiments on the small object detection datasets (SODAs) and the Small-DOTA dataset, constructed by filtering and downsampling DOTA to match SODA's characteristics, demonstrate that our method achieves state-of-the-art performance.

The remainder of this article is organized as follows. Section II reviews related small-object detection and the SAM in remote sensing. Section III elaborates on the details of the proposed RNP-OSOD framework. Section IV presents the experimental results and analysis. Finally, Section V concludes this article.

II. Related Work

A. Small-Object Detection in Remote Sensing Images

Detecting small objects in remote sensing images is a persistent challenge. These objects occupy a minimal number of pixels, and as deep neural networks grow in depth, their feature responses progressively weaken, complicating subsequent detection. To counteract this, researchers have developed numerous methods to amplify the features of small objects. These techniques can be broadly classified into three categories: object detail attention, object context modeling, and high-quality positive sample reselection. For object detail attention method, Li et al. [17] introduced a cross-layer attention network that uses a bidirectional feature pyramid to enrich contextual information, integrated with a module to capture nonlocal associations and strengthen feature representation. Yang et al. [18] designed an attention-based soft-threshold filtering module that uses a shrinkage attention mechanism to boost small-object semantics during upsampling and reduce feature redundancy and a feature-filtering pyramid to suppress redundant features and improve the semantic accuracy of small objects in high-resolution feature maps. Song et al. [19] proposed a semantic-driven dual attention branch, which is fused with spatial-level features to capture discriminative cues. The incorporation of boundary-sensitive supervision mitigates the loss of contour information in the fusion stage, strengthening object representational capacity. Shen et al. [20] innovatively treated the outputs of feature pyramid network (FPN) [21] as a temporal sequence, applying an object-focusing mechanism to emphasize small objects by providing both multiresolution features and surrounding context.

As for the object context modeling method, FS-SSD [22] incorporated spatial context analysis by computing interclass and intraclass distances between object instances, which is applied during object with low confidence redetection to improve the accuracy of multiclass small-object detection. Xu et al. [23] proposed a prior-guided context fusion network for small-object detection. The network decoupled scene contextual information through three components and dynamically integrates texture, boundary, and background priors to enhance localization and classification, which provided contextual support for small objects with weak textures or limited features and helped them stand out against complex backgrounds. Zhao et al. [24] proposed a scene contextualized detection network that decoupled scene contextual information via a scene classification subnetwork to explore relationships between objects and their surroundings. The network included a lightweight scene context-guided fusion module to incorporate context around dense tiny objects, and a foreground enhancement module that used scene information to suppress background and enhance foreground features. Zhou et al. [25] leveraged contextual information to attack synthetic aperture radar (SAR) target detectors, focusing on destroying the inherent interactions between targets and their surroundings, which are represented as strong pixel-level and contextual dependencies in SAR images.

For a high-quality positive sample reselection method, Wang et al. [26] modeled the object bounding box as a 2-D Gaussian distribution, replacing the intersection-over-union (IoU)-based metric with the proposed normalized Wasserstein distance for positive and negative sample assignments. This approach effectively alleviates the sensitivity of the IoU metrics to the location deviation of the small objects. Yuan et al. [5] utilized a coarse-to-fine region proposal network (RPN) that employed a dynamic anchor selection strategy and cascade regression to ensure sufficient and high-quality positive sample assignment. This approach is complemented by a feature imitation branch that enhances the representation of these difficult instances through supervised contrastive learning. Li et al. [27] introduced a dual weighting paradigm that assigned positive weights based on the consistency between classification and localization scores. Conversely, negative weights were independently determined by decomposing the weight into the probability of a sample being negative and its importance conditioned on that status.

Currently, some methods employed pseudo-masks to assist detection. SCRDet [28] employed coarse semantic segmentation ground truth (GT) by filling all bounding boxes in an image to supervise the feature denoising process, thereby enhancing the detection of small and densely packed objects. Similarly, Zhang et al. [29] builded upon this by separating semantic GTs for different object classes, obtaining spatially explicit attention maps with clear category information through

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feature supervision. While these approaches leveraged pseudo-masks to enhance object detection capabilities, their mask GTs lacked sufficient granularity and did not perform instance-level supervision. In the proposed RNP-OSOD, not only are finer pseudo-masks obtained, but the SAIF loss enables feature-level and instance-level supervision during network training.

Beyond object detection, advanced representation learning and context-driven strategies have also been successfully adapted to other challenging remote sensing tasks, including cross-domain road extraction [30], [31], target tracking in data-scarce hyperspectral video [32], [33], and complex scene interpretation using multimodal data or neural architecture search [34].

B. SAM in Remote Sensing

Visual foundation models represent a transformative shift in computer vision by providing versatile, large-scale neural networks that can be adapted to a wide array of downstream tasks such as object detection, image segmentation, and creative generation without requiring extensive task-specific retraining [35], [36], [37], [38]. They demonstrate remarkable zero-shot performance, maintaining high accuracy even when applied to specialized domains [39], [40], [41] despite being trained predominantly on large-scale natural image datasets. As a visual foundation model, the SAM [16] has recently emerged as a focal point in computer vision due to its powerful segmentation capabilities. SAM can precisely identify the position and outline of objects, allowing clear separation between foreground and background elements.

Recently, in remote sensing field, several studies have explored SAM's potential. SAM_MLoRAF framework [42] addressed generalization issues in urban object extraction by injecting parallel low-rank adaptation (LoRA) structures into the SAM encoder and employing a pyramid decoder to integrate multilevel information. Ren et al. [43] evaluated SAM's zero-shot performance on various benchmark tasks for overhead imagery, revealing that while SAM often generalizes well, it also exhibits systematic failures due to the unique characteristics of remote sensing objects and scenes. To address the laborious task of manual annotation in planetary science, Julka and Granitzer [44] developed a method that uses SAM with strategic prompts to generate a small set of high-quality annotations for training a domain-specific decoder. Moving beyond standard prompt mechanisms, Osco et al. [45] investigated the application of the SAM to remote sensing imagery and proposed an automated technique that integrated text prompting for grounding DINO [46] to derive bounding boxes compatible with SAM with one-shot training to boost its performance. Chen et al. [47] designed a prompter to adaptively generate prompts for improving the performance of SAM in instance segmentation. ROS-SAM [48] proposed a SAM based on LoRA fine-tuning to improve generalization capabilities in different remote sensing domains. During inference, this method obtains high-quality interactive segmentation of moving objects by sliding the input object box.

For obtaining segmentation labels of remote sensing objects, Wang et al. [6] proposed a pipeline to construct large-scale segmentation datasets using SAM with input of detection annotations in existing object detection, including point, bounding box, and mask information. Although SAM exhibits exceptional zero-shot segmentation performance when provided with high-quality prompts, its reliance on horizontal bounding boxes as input prompts presents a significant limitation when leveraging existing rotated bounding box annotations, which are prevalent in the remote sensing domain. Directly using the minimum circumscribed horizontal bounding box of a rotated object as a prompt inevitably incorporates substantial background pixels, introducing noise that often leads to degraded segmentation boundaries and erroneous inclusion of adjacent objects.

To address this fundamental mismatch between prompt geometry and object orientation, the RNP-SAM strategy effectively decouples the object from its background through a spatial and geometric normalization process.

III. Proposed RNP-OSOD Framework

As illustrated in Fig. 2, the proposed RNP-OSOD framework is designed in two distinct phases. The first mask generation phase aims to generate pixelwise pseudo-segmentation labels to supply auxiliary supervision to the detection network based on the proposed RNP-SAM. The second mask supervising object detection phase utilizes the pseudo-segmentation labels along with the SAIF loss to supervise network training.

A. Overview of the Proposed RNP-OSOD Framework

The proposed RNP-OSOD framework includes two phases, i.e., mask generation and mask supervising object detection, and is proposed to incorporate pixelwise pseudo-labels to supply auxiliary supervision to the detection network, strengthening the model's ability to represent fine-grained object structural details under a complex background.

To facilitate high-quality instance-aware and semantically meaningful mask generation, RNP-SAM first extracts image patches containing objects. It, then, applies rotational correction and scale augmentation to these patches before feeding the normalized inputs into SAM. This process yields reliable mask annotations for supervised learning. Given an input image I and all object annotations $\mathcal{B} = \{RBB_i\}_{i=1}^{N}$, the process iterates over each object to generate normalized inputs and obtain corresponding masks

$$Y_{gt} = \text{RNP-SAM}\,(I, \mathcal{B}) \tag{1}$$

where Y_{gt} is the generated segmentation label for the entire image. This process is detailed in Section III-B.

The generated masks are subsequently decomposed into two types of pseudo-labels: instance segmentation labels and semantic segmentation labels. The instance segmentation pseudo-labels are utilized to supervise an auxiliary instance segmentation head, thereby improving the model's ability to discriminate individual objects. Meanwhile, the semantic segmentation pseudo-labels guide an object-scale awareness feature reconstruction process through SAS loss. These two types of loss constitute SAIF loss

$$L_{SAIF} = L_{SAS}\left(\hat{Y}_{seg}, Y_{gt}\right) + L_{ins}\left(\hat{Y}_{ins}, Y_{gt}\right). \tag{2}$$

The SAIF loss is detailed in Section III-C.

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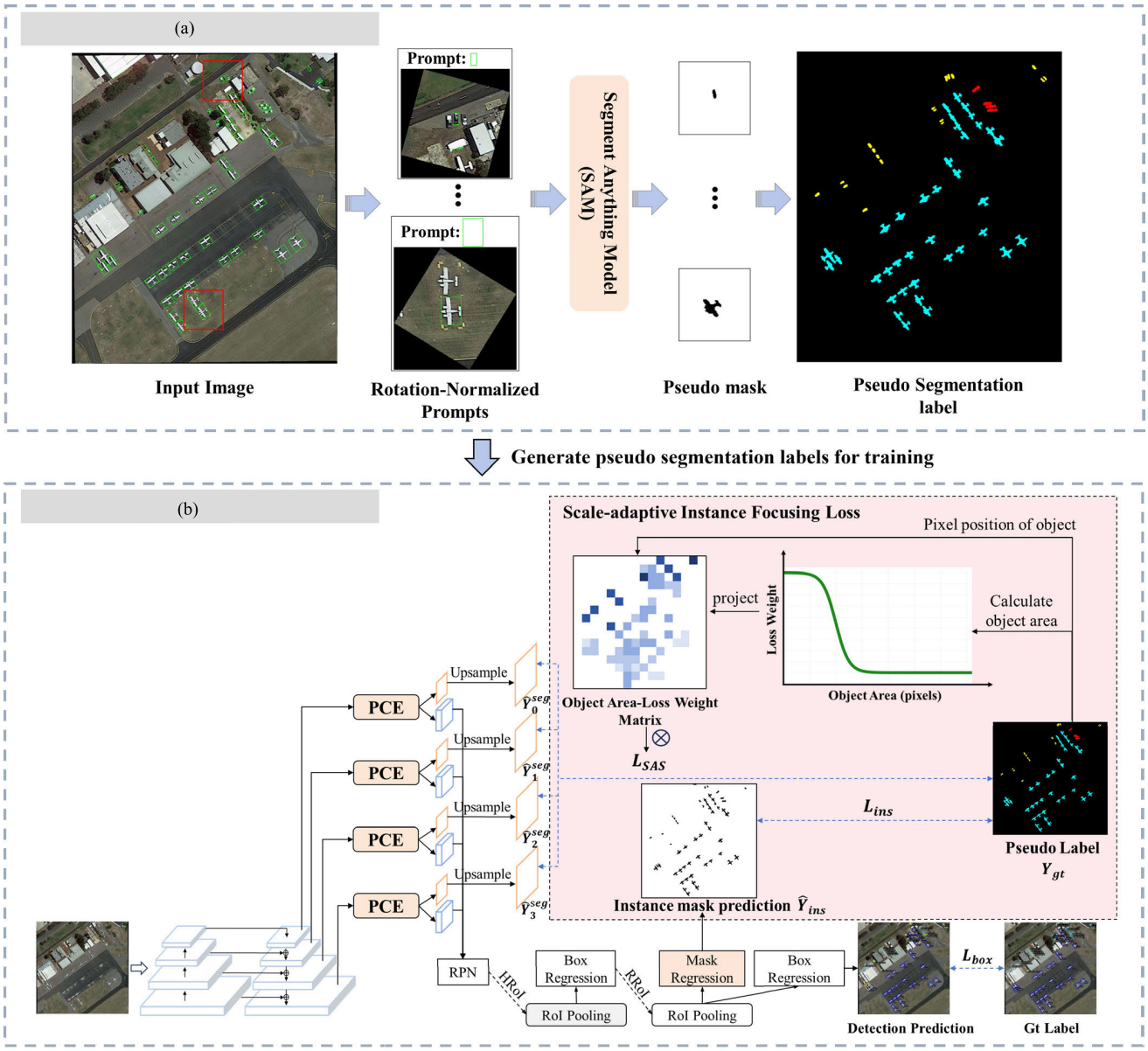


Fig. 2. Overall framework of the proposed RNP-OSOD. The framework comprises two phases. In the first RNP-SAM phase, both the object slice and rotated bounding box are simultaneously corrected to a horizontal orientation before being input into SAM, yielding a precise object mask. All masks are then restored to their original positions to generate pseudo-segmentation labels. In the second object detection phase, the pseudo-segmentation label serves as the GT Y_{gt} to supervise both the feature-predicted semantic mask $\hat{Y}_i^{seg}, i \in [0, 3]$ and the instance mask $\hat{Y}_{ins}$ derived from the instance segmentation head prediction. L_{SAS} and L_{ins} correspond to the respective losses for these two components. (a) Mask generation by RNP-SAM. (b) Mask supervising object detection.

B. Rotation-Normalized Prompt for SAM

As depicted in Fig. 2, RNP-SAM begins by processing each rotated bounding box annotation, defined as $RBB_i = (cx_i, cy_i, w_i, h_i, \theta_i)$, where (cx_i, cy_i) denotes the center coordinates, (w_i, h_i) represents the width and height, and θ_i signifies the rotation angle relative to the horizontal axis.

The initial processing stage involves extracting a localized image region centered precisely on the RBB_i centroid (cx_i, cy_i) with carefully controlled contextual boundaries. The cropping dimensions are determined using a fixed context margin d that defines the extent of the surrounding background to retain

$$\begin{aligned} \text{crop}_{\text{width}} &= w_i + 2d \\ \text{crop}_{\text{height}} &= h_i + 2d. \end{aligned} \tag{3}$$

This formulation ensures dimensional consistency while maintaining appropriate contextual information around the object. The resulting cropped patch I_i^{crop} preserves the essential visual context necessary for accurate segmentation while eliminating extraneous background elements that could potentially degrade the performance of SAM.

Next, to improve segmentation precision, particularly for small objects, we introduce an amplification step before rotation correction. The cropped patch I_i^{crop} is resized using nearest-neighbor interpolation with a fixed scaling factor $s > 1$

$$I_i^{amp} = \text{resize}_{\text{nearest}}\left(I_i^{crop}, s \cdot \text{crop}_{\text{width}}, s \cdot \text{crop}_{\text{height}}\right). \tag{4}$$

This preamplification approach enhances fine details before rotation transformation, minimizing interpolation artifacts during subsequent processing stages.

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Following amplification, the enhanced patch undergoes rotation correction to eliminate rotational variance. The transformation involves rotating I_i^{amp} by $-\theta_i^{\circ}$ around its center using an affine transformation. The affine transformation matrix R for rotating the amplified patch I_i^{amp} by angle θ_i around its center (cx_i, cy_i) is compactly defined by its key components. Let $a = \cos\theta_i$ and $b = \sin\theta_i$, and the transformation of a point (x, y) to (x', y') is given by

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} \tag{5}$$

where the affine transformation matrix R is

$$R = \begin{bmatrix} a & b & t_x \\ -b & a & t_y \end{bmatrix} \tag{6}$$

where the translation vector $\begin{bmatrix} t_x \\ t_y \end{bmatrix}$ ensures rotation about the specified center point and is calculated as

$$t_x = (1-a)\cdot cx_i + b \cdot cy_i, \quad t_y = -b \cdot cx_i + (1-a)\cdot cy_i. \tag{7}$$

This transformation produces a normalized image patch I_i^{norm}, where the original rotated bounding box is converted to a horizontal bounding box $HBB_i^{norm} = (cx_i, cy_i, w_i, h_i)$, effectively aligning the object's principal axis with the horizontal orientation. Direct rotation of small objects introduces severe aliasing and interpolation artifacts, so the amplification step is placed before the rotation step in the entire pipeline. Amplifying local regions before applying an affine rotation enhances sampling resolution, minimizing structural information loss caused by interpolation.

The normalized image patch I_i^{norm} and its corresponding horizontal bounding box HBB_i^{norm} are then presented to SAM for segmentation

$$M_i^{norm} = \text{SAM}\left(I_i^{norm}, HBB_i^{norm}\right). \tag{8}$$

This operation occurs in a geometrically normalized space where the object appears in its canonical orientation, free from rotational artifacts. The consistent presentation format enables SAM to leverage its learned priors more effectively, resulting in improved segmentation accuracy and boundary precision.

The generated mask M_i^{norm} undergoes a series of inverse transformations to restore its original orientation and size. First, the mask undergoes an inverse geometric transformation to restore its original orientation. The inverse rotation is applied using the transformation matrix R^{-1} with angle $+\theta_i$, and the transformation of a point (x, y) to (x', y') is given by

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_{xinv} \\ t_{yinv} \end{bmatrix} \tag{9}$$

where the affine transformation matrix R^{-1} is

$$R^{-1} = \begin{bmatrix} a & -b & t_{xinv} \\ b & a & t_{yinv} \end{bmatrix} \tag{10}$$

where the translation vector $\begin{bmatrix} t_{xinv} \\ t_{yinv} \end{bmatrix}$ ensures rotation about the specified center point and is calculated as

$$\begin{aligned} t_{xinv} &= (1-a)\cdot cx_i - b \cdot cy_i \\ t_{yinv} &= b \cdot cx_i + (1-a)\cdot cy_i. \end{aligned} \tag{11}$$

The amplified object mask M_i^{amp} is obtained through inverse rotation transformation

$$M_i^{amp} = R^{-1}\left(M_i^{norm}\right) = R^{-1}\left(+\theta_i\right) \cdot M_i^{norm}. \tag{12}$$

The rotated mask then undergoes downscaling by a factor $1/s$ using nearest-neighbor interpolation to revert the amplification

$$M_i^{rot} = \text{resize}_{\text{nearest}}\left(M_i^{amp}, \text{crop}_{\text{width}}, \text{crop}_{\text{width}}\right). \tag{13}$$

Algorithm 1 Rotation-Normalized Prompting for SAM

input : Original image I, list of rotated boxes $\mathcal{B} = [RBB_1, ..., RBB_N]$, context size d
output: Segmentation label Y_{gt} $Y_{gt} \leftarrow \text{zeros}(\text{shape}(I))$
for *each* $RBB_i = (cx_i, cy_i, w_i, h_i, \theta_i)$ *in* $\mathcal{B}$ **do**
  $crop_w \leftarrow w_i + 2d$;
  $crop_h \leftarrow h_i + 2d$;
  $I_i^{crop} \leftarrow \text{crop}(I, \text{center} = (cx_i, cy_i), \text{size} = (crop_w, crop_h))$;
  $I_i^{amp} \leftarrow \text{resize}_{\text{nearest}}(I_i^{crop}, s \times crop_w, s \times crop_h)$;
  $I^{norm}i \leftarrow R(I^{amp}i, -\theta_i)$;
  $HBB_i^{norm} \leftarrow (cx_i, cy_i, w_i, h_i)$;
  $M_i^{norm} \leftarrow \text{SAM}(I_i^{norm}, HBB_i^{norm})$;
  $M_i^{amp} \leftarrow R^{-1}(M_i^{norm}, +\theta_i)$;
  $M_i^{rot} \leftarrow \text{resize}_{\text{nearest}}(M_i^{amp}, \frac{1}{s} \times crop_w, \frac{1}{s} \times crop_h)$;
  $Y_{gt} \leftarrow \text{place}(Y_{gt}, M_i^{rot}, \text{at} = (cx_i, cy_i))$
end

The final processing stage involves integrating the transformed mask M_i^{rot} back into the global image coordinate system. The mask is precisely positioned using the original centroid coordinates (cx_i, cy_i), with careful attention to spatial alignment and coordinate mapping. Since RNP-SAM processes objects individually, overlapping regions are minimal. The segmentation map is reconstructed by accumulating values back to the original object locations. Background pixels are represented as 0, while foreground pixels use 255 minus the label id. If overlapping target pixels occur during accumulation, the pixel value exceeds 255. When this situation is detected, the pixel is directly assigned the value corresponding to the current target category being accumulated. The integration process employs a composite operation that aggregates all individual object masks into a comprehensive segmentation label Y_{gt}, ensuring proper handling of potential overlaps between adjacent objects. The complete RNP-SAM algorithm is formally specified as Algorithm 1.

This comprehensive approach effectively addresses the fundamental challenges associated with using rotated bounding boxes as prompts for SAM, providing a robust framework for generating high-quality segmentation masks from existing RBB annotations while minimizing background interference and maximizing segmentation accuracy.

C. SAIF Loss

SAIF loss is proposed to sharpen the model's sensitivity to small objects and their structural details. This is a novel composite loss function that consists of two synergistic components: an SAS loss that adaptively reweights small objects

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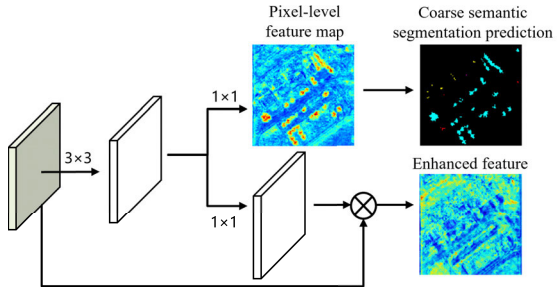


Fig. 3. Diagram of PCE module. It extracts pixel-level cues from the features and weights them into the original features.

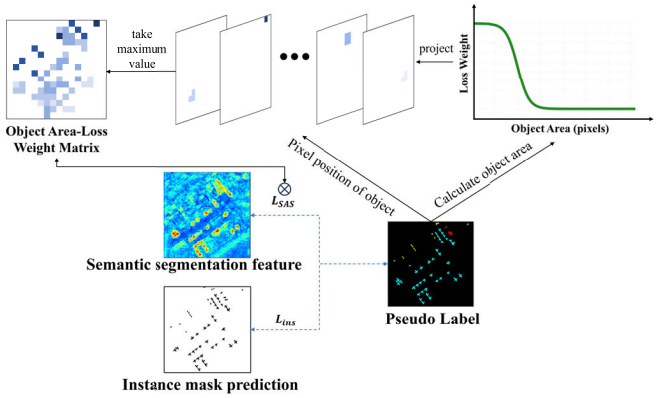


Fig. 4. Diagram of SAIF loss. It comprises two components: SAS loss, in which loss weights are generated according to the object area, and an instance segmentation loss decoupled from the detection loss.

based on their areas, and a detection-decoupled instance segmentation loss that enhances the model's ability to characterize object boundaries and shape information. Together, these components enable the model to dynamically focus on harder and smaller instances while improving localization accuracy and feature discriminativeness.

SAS loss is built upon a pixel-level cues extraction (PCE) module, which operates on each level of the feature pyramid. As shown in Fig. 3, for a given feature map P_i from the FPN, PCE applies a 3×3 convolution for contextual feature blending, followed by two parallel 1×1 convolutions. One branch outputs a semantic segmentation prediction $\hat{Y}_i^{seg}$, while the other produces a spatial attention map that highlights semantically salient regions. This attention map is then multiplied elementwise with the original feature P_i, effectively suppressing background noise and amplifying foreground responses, particularly for smaller objects.

As depicted in Fig. 4, SAS loss adaptively assigns weights to objects of varying scales based on their areas, and supervises semantic segmentation predictions $\hat{Y}_i^{seg}$ using pseudo-segmentation labels Y_{gt} generated by RNP-SAM. Here, we require a function that assigns greater loss weights to smaller objects, decreasing monotonically with object area. To ensure training stability, the function's range should remain relatively stable across different object scales. The sigmoid function exemplifies a monotonically decreasing function with a range in the (0, 1) interval. Therefore, the object area-loss weight function is designed as follows:

$$\omega(A) = \sigma(k \cdot (A - A_{\text{ref}})) = \frac{1}{1 + e^{k \cdot (A - A_{\text{ref}})}} \tag{14}$$

where $A_{\text{ref}} = 400$ is set according to the SODA [49] dataset's definition of "relatively small" objects, and the slope parameter k controls the degree of emphasis on objects smaller than A_{ref}. The threshold of $A_{\text{ref}} = 400$ is employed because smaller objects contain fewer pixels, making them more challenging to detect. Furthermore, such diminutive objects are relatively scarce within the dataset, leading to sample imbalance. This imbalance risks allowing the more numerous smaller objects to dominate during training. Consequently, a higher weighting must be assigned to them within the loss function. Formula 14 produces higher loss weights for smaller instances, effectively addressing the scale imbalance during training. The weight values are propagated to all pixels within each instance region, forming an object area-loss weight matrix W. For the segmentation label $Y_{gt} \in \mathbb{Z}^{H \times W}$ and a set of weights $\{\omega(A_i)\}_{i=1}^{K}$ for each of the K instances, the corresponding loss weight matrix $W \in \mathbb{R}^{H \times W}$ is

$$W = \sum_{i=1}^{K} \omega(A_i) \cdot M_i \tag{15}$$

where M_i is the binary indicator matrix for instance i, defined as

$$M_i(x, y) = \begin{cases} 1, & \text{if } Y_{gt}(x, y) = i \\ 0, & \text{otherwise.} \end{cases} \tag{16}$$

The SAS loss combines cross-entropy and multiclass Dice loss [50], aggregated across all FPN levels and modulated by W

$$L_{\text{SAS}} = \sum_{n=1}^{N} \left[-\frac{1}{P} \sum_{i=1}^{P} \sum_{k=1}^{C} Y_{gt}(i,k) \log\left(\hat{Y}_n^{seg}(i,k)\right) + 1 - \frac{1}{C} \sum_{k=1}^{C} \frac{2 \sum_{i=1}^{P} Y_{gt}(i,k)\ \hat{Y}_n^{seg}(i,k)}{\sum_{i=1}^{P} Y_{gt}(i,k) + \sum_{i=1}^{P} \hat{Y}_n^{seg}(i,k) + \epsilon} \right] \odot W \tag{17}$$

where $P = H \times W$ and $N = 4$ denote the number of FPN levels, and C represents the number of categories. This loss not only enhances the discrimination between foreground and background but also balances the quantitative differences among small objects at various scales.

The second component of SAIF loss introduces a detection-decoupled instance segmentation loss to stimulate the learning of fine-grained features such as contours and shape attributes. An additional segmentation head is attached to the detection network, producing per-instance mask predictions. This head is trained using pseudo-instance segmentation labels, leveraging pixelwise cross-entropy loss

$$L_{ins} = -\frac{1}{M^2} \sum_{i=1}^{M^2} \left(Y_{gt}(i) \log\left(\hat{Y}_{ins}(i)\right) + \left(1 - Y_{gt}(i)\right) \log\left(1 - \hat{Y}_{ins}(i)\right) \right) \tag{18}$$

where M denotes the mask output size for each RoI.

The overall SAIF loss is then given by

$$L_{\text{SAIF}} = L_{\text{SAS}} + \lambda L_{ins} \tag{19}$$

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where λ is the balanced factor of the two losses.

IV. Experiments and Analysis

A. Experimental Setup

1) Implementation Details: To rigorously evaluate the effectiveness of our proposed method, all original images in SODA-A [49] are first cropped into overlapping patches of size 800×800 pixels with a stride of 650 pixels, and similar to [49], these patches will be resized to 1200×1200 during training and testing. As for Small-DOTA, all original images are first cropped into overlapping patches of size 800×800 pixels with a stride of 650 pixels. The experiments are conducted using a single NVIDIA A6000 RTX GPU. We adopt a ResNet-50 backbone with an FPN to enhance multiscale feature representation, which is particularly beneficial for detecting extremely small objects. The network is trained using stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay of 0.0001. The learning rate is initially set to 0.01, and a batch size of 2 is used. To ensure adequate convergence, the model is trained for a total of 12 epochs on SODA-A and 72 epochs on Small-DOTA. During training, random horizontal and vertical flips are applied as data augmentation to improve model robustness against variations in object orientation. During inference, low-confidence predictions are filtered using a threshold of 0.05, and nonmaximum suppression (NMS) with an IoU threshold of 0.5 is applied to remove redundant bounding boxes, while the maximum number of predicted boxes per image is limited to 3000. In the loss function, the balance factor of SAIF loss is set to $\lambda = 0.5$ to maintain a proper tradeoff between SAS loss and instance segmentation loss. This configuration, combined with the adaptive resizing of the Small-DOTA dataset and multiscale FPN features, enables effective detection of small objects in a complex background.

2) Evaluation Metrics: To quantitatively assess the performance of our proposed detection method, we adopt the widely used average precision (AP) metric, following the evaluation protocol of MS COCO [51], but tailored to focus on small objects in remote sensing imagery. The AP metric provides a comprehensive measure of both precision and recall across varying IoU thresholds, and is formally defined as

$$\mathrm{AP} = \frac{1}{T}\sum_{i=1}^{T} P(r_i) \tag{20}$$

where T represents the number of discrete IoU thresholds, r_i denotes the recall at the ith threshold, and $P(r_i)$ corresponds to the precision measured at that recall level. Precision and recall are computed as

$$\begin{aligned} \mathrm{Precision} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ \mathrm{Recall} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \end{aligned} \tag{21}$$

where TP (true positives) represents the number of correctly detected objects, FP (false positives) denotes the number of incorrectly detected objects, and FN (false negatives) refers to the number of missed objects. The IoU between predicted and GT boxes is used to determine whether a detection is counted as TP.

TABLE I
Small-Object Area Subsets and Corresponding Object Size Ranges [49]

Subset	Object Area (pixels)
Extremely small (es)	(0, 144]
Relatively small (rs)	(144, 400]
Generally small (gs)	(400, 1024]

In our experiments, we employ the same AP-based metrics as SODA-A to capture performance across different IoU thresholds and object sizes [49].

1) *Overall AP:* Computed as the mean AP over ten IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05, focusing exclusively on small objects.
2) *AP_{50} and AP_{75}:* Calculated at single IoU thresholds of 0.5 and 0.75, respectively. These metrics emphasize detection performance for objects with limited size.
3) *Area-Specific AP:* To evaluate performance across varying small-object scales, we divide the small objects into three area subsets: *extremely small (eS)*, *relatively small (rS)*, and *generally small (gS)*. The corresponding AP metrics are denoted as AP_{eS}, AP_{rS}, and AP_{gS}.

Table I summarizes the small-object area ranges used to categorize objects for the area-specific AP metrics.

Through comparing both overall and area-specific AP scores, along with AP_{50} and AP_{75}, we provide a detailed evaluation of our method's ability to detect small objects under varying scales and strict IoU criteria.

B. Experiments 1: SODA-A Dataset

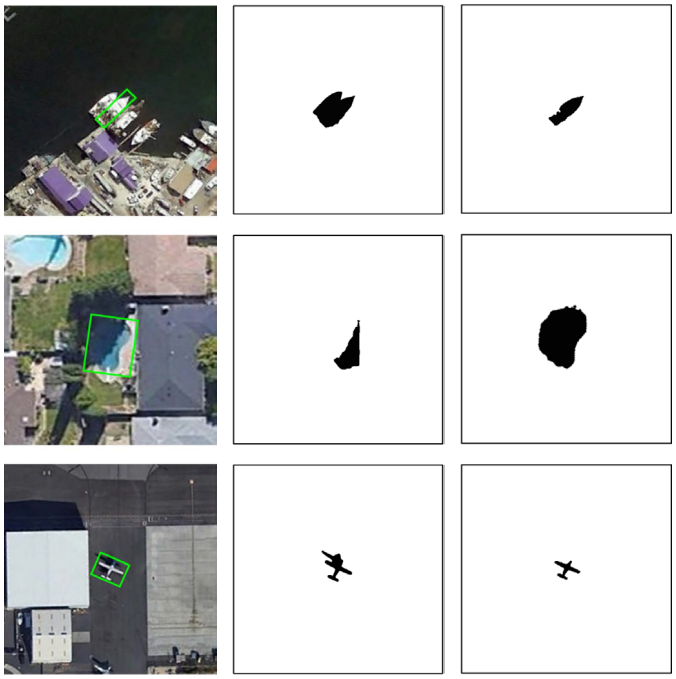
1) SODA-A Dataset: SODA-A [49] comprises 2513 aerial images with a total of 872 069 annotated objects, each represented by an oriented bounding box. The dataset covers nine object categories: *airplane, helicopter, small vehicle, large vehicle, ship, container, storage tank, swimming pool, and windmill.* Objects within SODA-A may appear in any rotation and exhibit substantial variations in spatial density, averaging approximately 350 instances per image. As a benchmark specifically designed for detecting small objects, SODA consists of instances that are predominantly minute, typically ranging from 10 to 30 pixels in size. Unlike typical object detection datasets, it incorporates extensive "ignore regions" to exclude examples that exceed size thresholds or present ambiguous visibility due to severe occlusion or imaging artifacts such as lens flare. This strategy encourages models to concentrate on learning from meaningful small objects. Instances in SODA are categorized based on area into small and normal, with the small class further refined into three subcategories: eS, rS, and gS.

2) Availability for Generated Pseudo-Label: Fig. 5 visualizes the segmentation pseudo-labels generated by RNP-SAM on SODA-A. The same scene and target as in Fig. 1(b) were selected for segmentation result comparison. Results demonstrate that RNP-SAM maintains precise segmentation performance for small targets under three complex back-

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TABLE II
COMPARED WITH OTHER METHODS ON SODA-A. RED REPRESENTS THE BEST PERFORMANCE. BLUE REPRESENTS THE SECOND-BEST PERFORMANCE

Method	AP	AP_{50}	AP_{75}	AP_{es}	AP_{rs}	AP_{gs}	AP_N
Rotated FasterRCNN (2017) [7]	33.3	70.6	25.5	13.0	28.3	43.0	35.2
Oriented R-CNN (2021) [52]	35.1	70.7	30.0	14.0	29.9	44.7	37.4
Rotated RepPoints (2019) [53]	15.3	36.6	9.8	4.5	14.3	17.8	17.2
Oriented RepPoints (2022) [54]	32.8	70.4	25.1	13.9	29.6	41.2	36.8
ReDet (2021) [55]	35.8	72.0	30.0	13.8	30.6	45.8	38.1
RoI Transformer (2019) [56]	36.4	72.8	31.0	15.1	31.1	46.2	38.9
Rotated RetinaNet (2020) [57]	24.8	60.5	13.9	9.1	20.4	32.5	26.1
Gliding Vertex (2021) [58]	31.8	70.6	22.6	12.4	27.4	41.0	33.4
S2A-Net (2022) [59]	14.2	50.5	1.2	5.4	11.6	16.7	15.6
Rotated FCOS (2019) [60]	33.7	70.9	26.7	12.5	28.3	44.2	34.5
Rotated ATSS (2020) [61]	32.8	69.5	25.7	12.0	27.5	43.3	33.1
CSL (2020) [62]	28.8	61.6	22.1	8.6	23.5	40.6	31.1
R3Det (2021) [63]	28.4	66.5	17.7	10.4	24.2	35.7	30.2
CFINet (2023) [5]	34.4	73.1	26.1	13.5	29.3	44.0	35.9
RNP-OSOD(proposed)	37.2(+0.8)	73.2(+0.1)	32.7(+1.7)	15.5(+0.4)	31.4(+0.3)	47.6(+1.4)	39.7(+0.8)



(a) (b) (c)

Fig. 5. Visualization comparison for generated pseudo-label on SODA-A between Wang et al. [6] and RNP-SAM, corresponding to the example in Fig. 1(b). (a) Object slice. (b) Object mask of Wang et al. [6]. (c) Object mask of the proposed RNP-SAM.

ground conditions within the SODA-A. Compared to the approach of directly using the minimum bounding rectangle from the rotated box as the SAM prompt, RNP effectively mitigates the effects of densely arranged targets, tree occlusions, and target shadows. This yields high-quality pixel-level pseudo-labels that can finely describe the appearance of small objects.

3) Comparison With Other Methods: Table II presents a comparison of detection results between the proposed RNP-OSOD and several state-of-the-art methods on the SODA-A dataset. The methods selected represent leading approaches in the field of oriented object detection. Two-stage methods include Rotated Faster RCNN [7], Oriented RCNN [52], Rotated RepPoints [53], Oriented RepPoints [54], ReDet [55], and RoI Transformer [56]; while single-stage methods include Rotated RetinaNet [57], Gliding Vertex [58], S2A-Net [59], Rotated FCOS [60], Rotated ATSS [61], CSL [62], and R3Det [63]. The results show that the proposed RNP-OSOD achieves the highest performance across all evaluation metrics. The proposed RNP-OSOD achieves an AP of 37.2%, surpassing the AP of the highest-accuracy method, RoI Transformer (also the baseline method), by 0.8%. For the three small-object detection metrics, AP_{es} is 0.4% higher, AP_{rs} is 0.3% higher, and AP_{gs} is 1.4% higher than the highest-accuracy method, RoI Transformer. We also highlighted the gains compared with the second-best metrics in Table II to clearly demonstrate the improvement. It can be obscured that although the AP improves by only 0.8%, the more stricter evaluation metric AP_{75} shows an increase of 1.7%. This demonstrates that the proposed RNP-OSOD exhibits robust performance in the SODA-A dataset.

Fig. 6 shows the comparison of detection results across different models for several scenes in the SODA-A dataset. The red dashed regions are magnified to clearly illustrate performance differences between methods. First, in scenes densely populated with large vehicle objects, the proposed RNP-OSOD maintains optimal performance on the SODA dataset. Second, when objects with high similarity to objects (such as road signs and large vehicles) appear in the background, both RNP-OSOD and baseline methods effectively suppress interference without generating any false alarms. Additionally, in scenarios where objects are extremely small or barely discernible, nearly blending into the background, the proposed RNP-OSOD consistently delivers high performance among all methods. For example, in Fig. 6 depicting aircraft at an airport, only the proposed RNP-OSOD detects some of these objects, while other methods fail to perceive their presence. Fig. 7 presents the object detection results and corresponding instance segmentation results of RNP-OSOD on the SODA-A. It is evident that the segmentation results for the head region exhibit a high degree of reliability. Whether dealing with small-object contours against complex backgrounds or objects with highly intricate structures such

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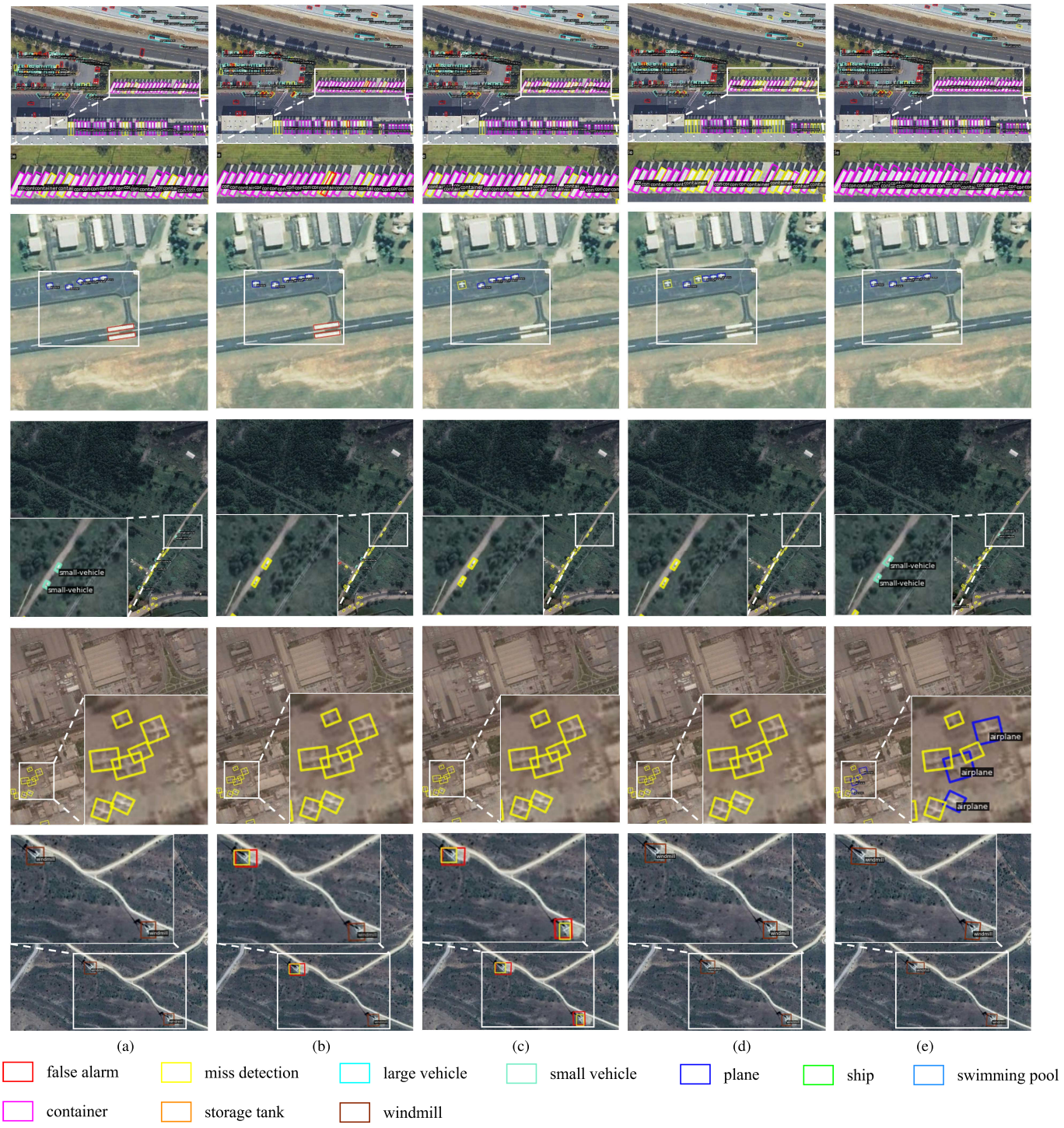


Fig. 6. Visualization of comparison with other methods on SODA-A. The red box represents false alarm (FP), and the yellow box represents missing detection (TN). (a) Oriented RCNN. (b) Gliding Vertex. (c) RoI Transformer. (d) CFINet. (e) RNP-OSOD (proposed).

as the windmill, the proposed RNP-OSOD achieves a precise segmentation effect.

To compare RNP-OSOD with other combinations of rotating detection and instance segmentation, the object detection heads of instance segmentation methods—MaskRCNN [64], Hybrid Task Cascade [65], PointRend [66], and SCNet [67]—were modified to perform regression predictions for rotating detection. Additionally, an instance segmentation head was added to the baseline rotating detection method, RoI Transformer. These methods were evaluated using detection metrics on SODA-A, with results shown in Table III. It is evident that the proposed RNP-OSOD significantly outperforms other combined approaches. Its AP exceeds the highest AP among the combined methods (Mask RoI Transformer) by 0.5%. Additionally, its small-object performance metrics show AP_{es} improved by 0.3%, AP_{rs} by 0.1%, and AP_{gs} by 0.9%.

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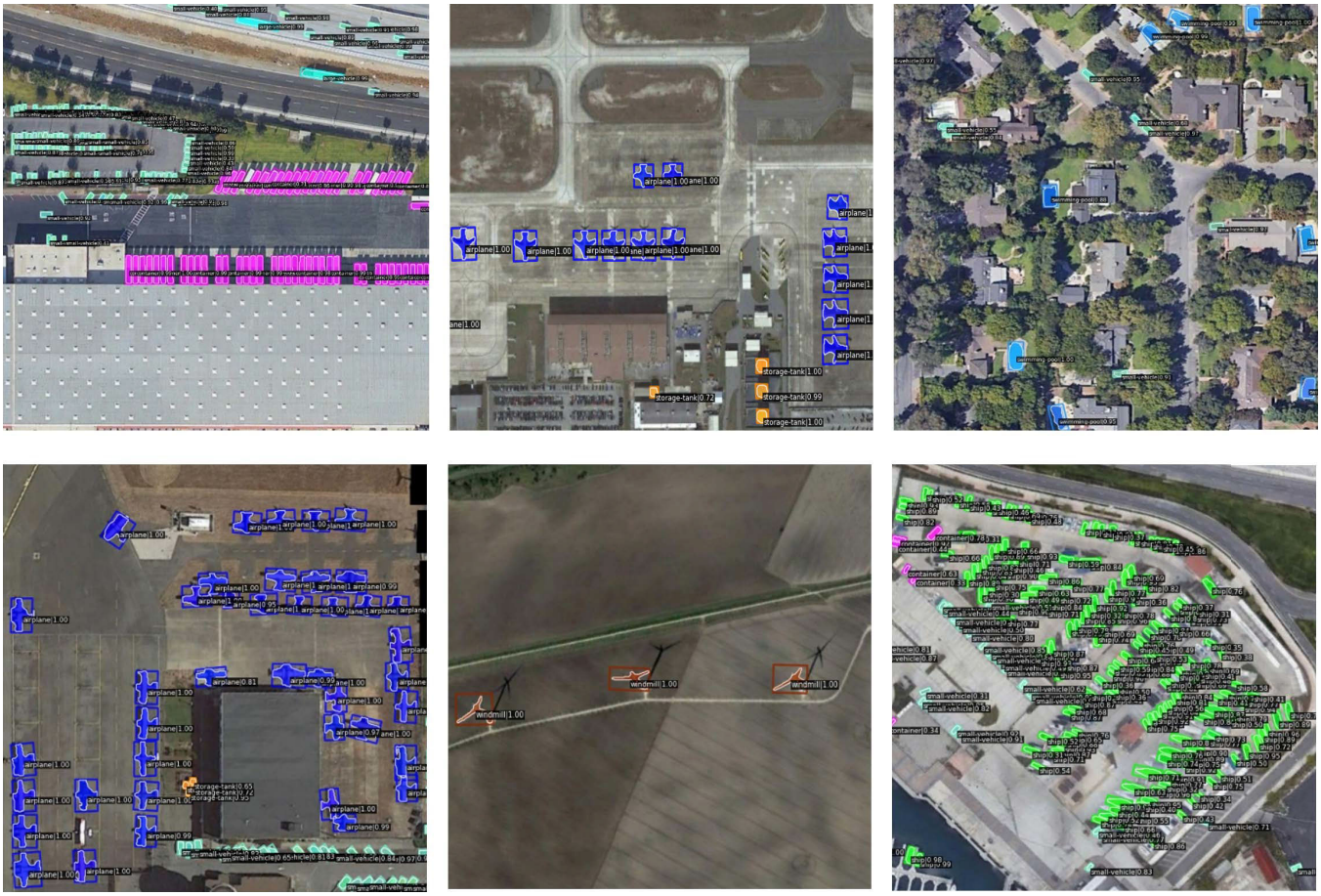


Fig. 7. Visualization of the proposed RNP-OSOD detection results and corresponding instance segmentation results on SODA-A.

TABLE III
RESULTS OF DIFFERENT ORIENTED OBJECT DETECTION METHODS WITH INSTANCE SEGMENTATION ON THE TEST SET OF THE SODA-A. RED REPRESENTS THE BEST PERFORMANCE. BLUE REPRESENTS THE SECOND-BEST PERFORMANCE

Method	AP	AP_{50}	AP_{75}	AP_{es}	AP_{rs}	AP_{gs}	AP_N
Rotated MaskRCNN	33.5	70.8	26.1	13.2	28.4	42.8	35.6
Rotated Hybrid Task Cascade	32.5	69.8	24.5	12.8	27.6	41.5	34.2
Rotated PointRend	28.4	64.7	18.9	10.2	24.5	35.7	30.4
Mask RoI transformer	36.7	73.0	31.9	15.2	31.3	46.7	39.3
Rotated SCNet	29.1	64.3	20.3	8.6	25.3	37.2	32.6
RNP-OSOD (proposed)	37.2	73.2	32.7	15.5	31.4	47.6	39.7

C. Experiment 2: Small-DOTA Dataset

1) Small-DOTA Dataset: To validate the performance of the proposed RNP-SAM for obtaining pseudo-segmentation labels of small objects, we select the DOTA dataset [68], an oriented object detection dataset with pixel-level annotations, and construct a specialized dataset according to SODA's characteristics [49] based on it, called Small-DOTA. The Small-DOTA dataset not only has pixel-level annotations but also exclusively contains small objects matching the SODA dataset's categories and size range.

DOTA-1.0 is a large-scale benchmark dataset for oriented object detection in aerial images. It contains 2806 high-resolution images ranging from 800 × 800 to 4000 × 4000 pixels, with over 188 000 annotated instances spanning 15 object categories. The objects in DOTA exhibit substantial variation in scale, orientation, and aspect ratio, reflecting the complexity of real-world remote sensing scenarios. While DOTA has been widely adopted for general object detection in remote sensing, it includes many medium and large objects that are not suitable for evaluating algorithms specifically on tiny objects.

To construct Small-DOTA, we first select a subset of object categories from DOTA-1.0 that are most relevant to small-object detection tasks. Only six categories are retained: *storage tank*, *large vehicle*, *small vehicle*, *plane*, *ship*, and *swimming pool*. This selection is motivated by the desire to align with other small-object benchmarks such as SODA, while maintaining sufficient diversity in object appearance, context, and scene types.

To evaluate the quality of pseudo-segmentation labels using the Small-DOTA dataset, GT mask annotations for the objects must be obtained. The isAID [69] dataset is a segmentation-annotated dataset constructed based on the DOTA dataset. Due

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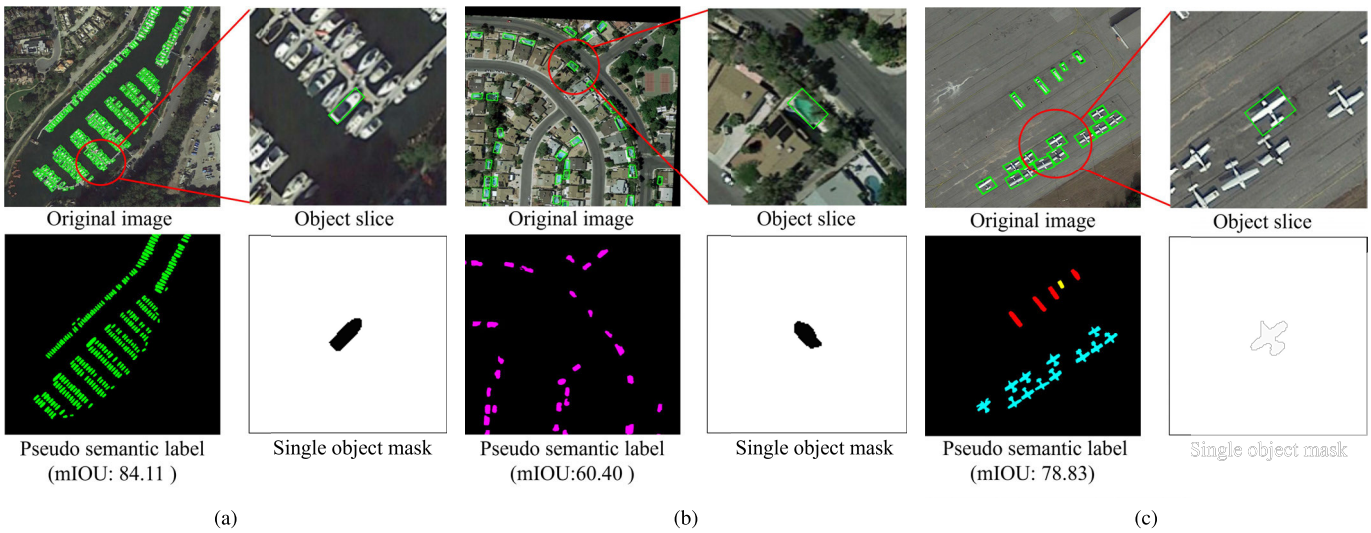


Fig. 8. Visualization for generated pseudo-label on Small-DOTA. (a)–(c) Three different sample images.

TABLE IV
DETECTION RESULT COMPARISON BETWEEN SUPERVISING BY RNP-SAM GENERATED LABELS AND GT LABELS

Method	Training annotation	AP	AP_{50}	AP_{75}	AP_{es}	AP_{rs}	AP_{gs}
RNP-OSOD (proposed)	GT	38.9	72.0	38.7	17.5	39.3	51.7
RNP-OSOD (proposed)	RNP-SAM	38.8	72.1	38.2	17.1	38.9	51.8

TABLE V
COMPARISON WITH OTHER METHODS ON SMALL-DOTA. RED REPRESENTS THE BEST PERFORMANCE. BLUE REPRESENTS THE SECOND-BEST PERFORMANCE

Method	AP	AP_{50}	AP_{75}	AP_{es}	AP_{rs}	AP_{gs}	Param.	GFLOPs
Rotated FasterRCNN (2017)	35.1	70.0	30.5	15.7	35.7	47.0	41.13	134.39
Oriented R-CNN (2021)	37.0	69.1	36.5	15.6	37.4	49.9	41.13	134.47
Rotated RepPoints (2019)	28.0	58.1	23.5	11.4	29.5	39.6	36.67	310.56
Oriented RepPoints (2022)	29.2	64.3	21.1	7.6	29.9	46.3	36.60	118.73
ReDet (2021)	33.3	61.9	32.5	6.8	36.0	51.3	31.58	59.78
RoI Transformer (2019)	38.0	70.5	37.5	16.6	38.3	50.2	55.06	148.32
Rotated RetinaNet (2020)	24.3	59.1	14.0	9.5	24.3	35.1	36.14	345.37
Gliding Vertex (2021)	34.3	69.9	28.7	14.7	35.2	44.8	41.13	134.39
S2A-Net (2022)	17.2	55.5	2.7	6.5	18.5	23.3	38.62	316.52
Rotated FCOS (2019)	32.8	65.4	29.2	7.1	32.0	50.2	31.90	126.13
Rotated ATSS (2020)	27.6	58.0	23.0	5.0	25.3	46.4	36.03	126.77
CSL (2020)	22.9	50.6	17.1	1.8	21.0	43.9	37.16	141.93
R3Det (2021)	25.2	56.1	19.3	3.2	24.0	43.5	41.70	202.46
CFINet (2023)	30.6	58.6	29.1	9.7	3.0	46.4	48.83	142.57
RNP-OSOD (proposed)	38.9(+0.9)	72.0(+1.5)	38.7(+1.2)	17.5(+0.9)	39.3(+1.0)	51.7(+0.4)	58.35	183.38
RNP-OSOD* (proposed)	38.8(+0.8)	71.9(+1.4)	38.7(+1.2)	17.2(+0.6)	38.9(+0.6)	51.6(+0.3)	58.35	183.38

"*" represents that RNP-SAM employs SAM2 to generate pseudo masks.

to differences in annotation volume between the two datasets, the objects within them must be aligned by removing instances annotated only with bounding boxes or only with instance masks. This transforms the Small-DOTA dataset into a small-object detection dataset with pixel-level annotations.

Next, we apply strict size constraints to ensure that all objects fall within the tiny-object range. Specifically, each object in Small-DOTA is resized such that its maximum dimension does not exceed 32 pixels, and its minimum dimension is no smaller than 3 pixels. This allows the dataset to focus exclusively on small objects, while preserving their relative proportions.

After resizing and category selection, the resulting Small-DOTA retains the dense object distributions and challenging spatial arrangements of the original DOTA images, while focusing entirely on tiny objects. Each image in Small-DOTA contains multiple small objects in various orientations and contexts, providing a rigorous benchmark for evaluating small-object detection algorithms in remote sensing scenarios.

The resizing and label filtering methods employed during the construction of the Small-DOTA dataset follow the SODA dataset methodology [49]. This approach increases the number of small objects while ensuring the data distribution remains unchanged. To validate the dataset's usability, we will publicly release the dataset construction script and scene-level segmentation scheme at https://github.com/cbq233333/RNP-OSOD. Small-DOTA provides a high-quality small-object detection

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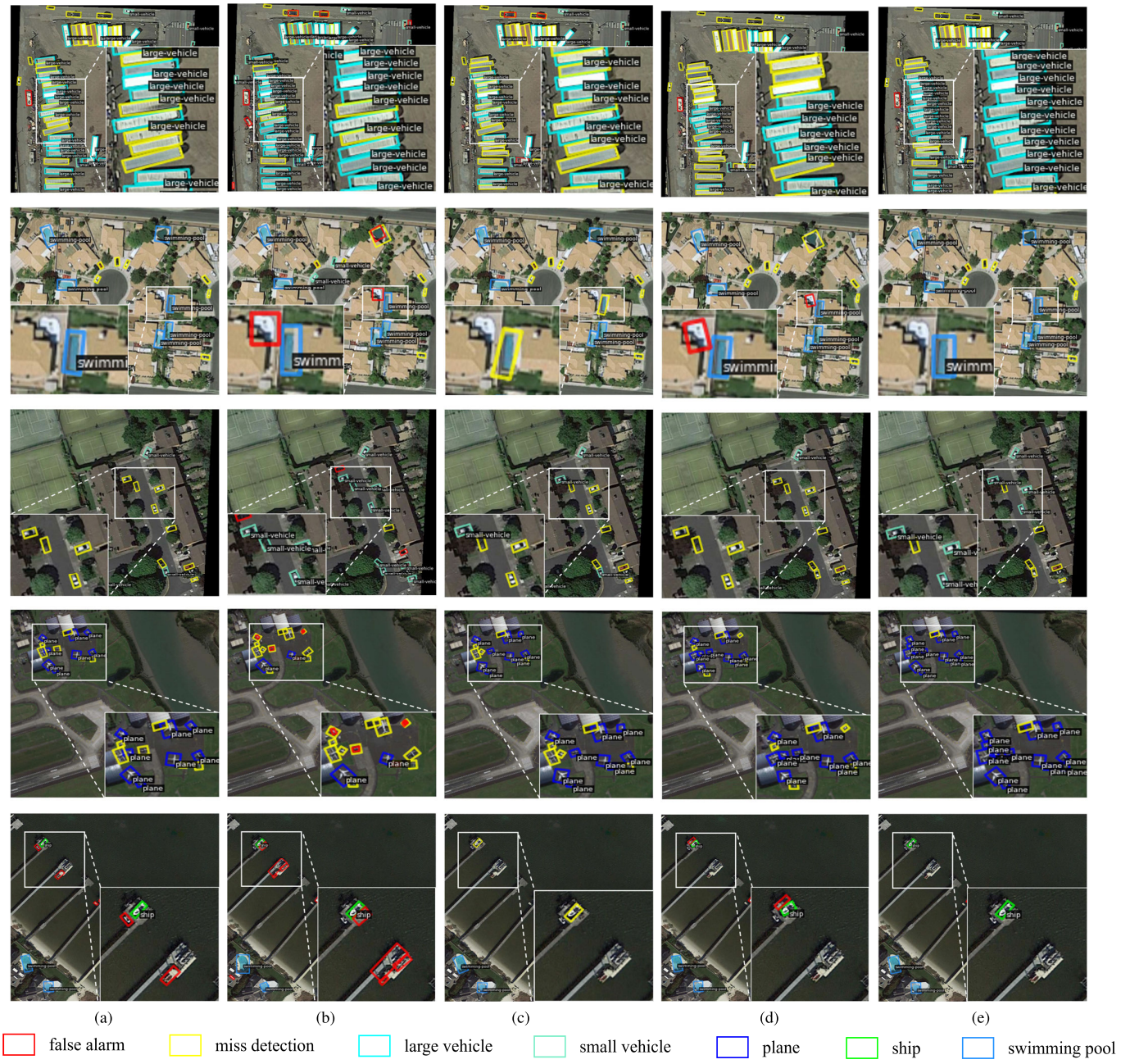


Fig. 9. Visualization of comparison with other methods on Small-DOTA. The results in the fourth row correspond to the example shown in Fig. 1(a). The red box represents false alarm (FP), and the yellow box represents missing detection (TN). (a) Oriented RCNN. (b) Gliding Vertex. (c) RoI Transformer. (d) CFINet. (e) RNP-OSOD (proposed).

benchmark with six representative object categories, strict size constraints, and adaptive resizing that preserves object scale ratios. It is particularly suitable for testing methods designed to handle dense, small, and oriented objects in aerial imagery.

2) Availability for Generated Pseudo-Label: Fig. 8 visualizes the segmentation pseudo-labels generated by RNP-SAM. The results show that RNP-SAM is also quite accurate in segmenting small-object details and produces very detailed semantic segmentation results for the entire scene. Furthermore, Table IV shows that the test metrics obtained by training the proposed method using GT and RNP-SAM-generated labels are comparable.

Fig. 8 demonstrates the performance of RNP-SAM-generated pseudo-mask labels in three distinct scenarios. For objects like ships in Fig. 8(a), where the proportion of pixels occupied by the object itself and its surrounding pixels is similar, RNP-SAM also preserves the object's shape and outline as much as possible, resulting in more distinguishable segmentation outcomes. For the swimming pool obscured by trees in Fig. 8(b), RNP-SAM can delineate the approximate shape of the object without being affected by the obstruction. It can be observed that despite some details of the small aircraft object being obscured within the complex background of the remote sensing image in Fig. 8(c), RNP-SAM still clearly segments the structural

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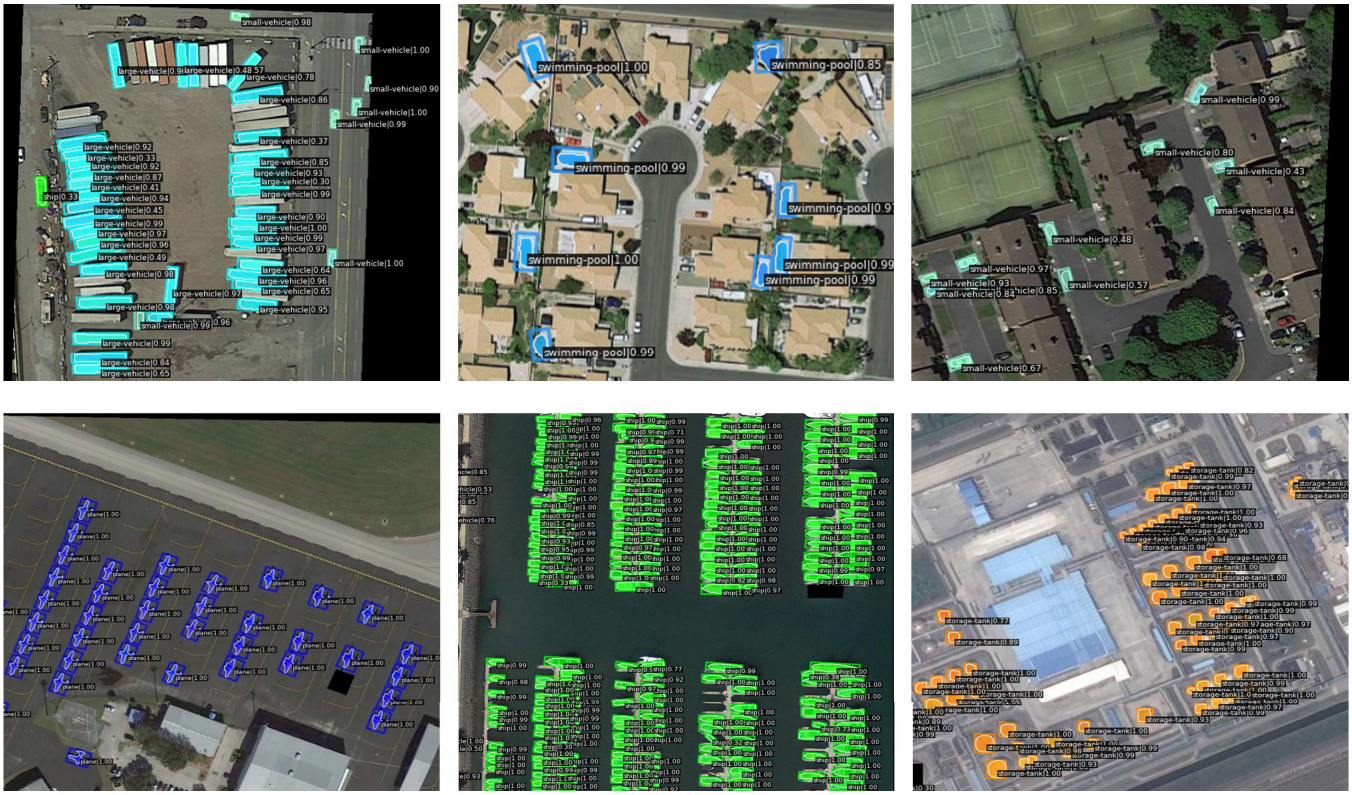


Fig. 10. Visualization of the proposed RNP-OSOD detection results and corresponding instance segmentation results on Small-DOTA.

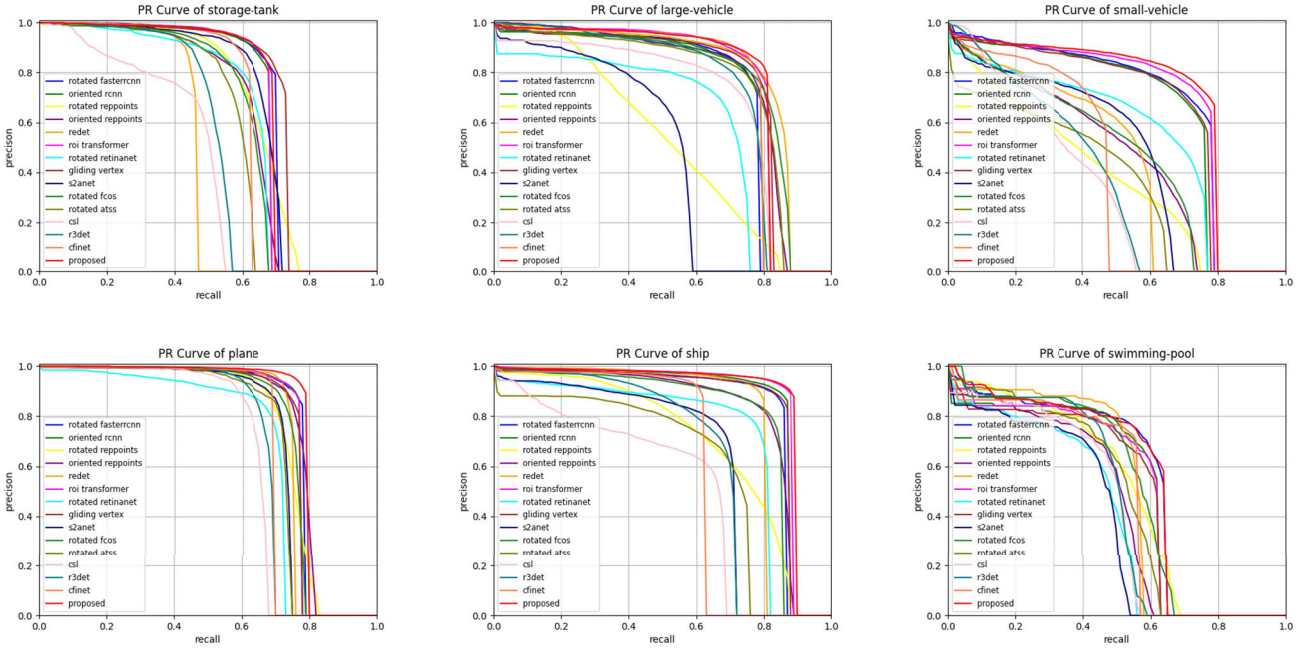


Fig. 11. PR curves of comparison with other methods on every class of Small-DOTA.

skeleton of the aircraft. Overall, the visualization results indicate that the pseudo-labels generated by RNP-SAM exhibit high reliability.

Table IV further demonstrates the detection results obtained by training the proposed network using both GT labels and pseudo-labels generated by RNP-SAM. The data indicate that the performance metrics trained with RNP-SAM pseudo-labels show minimal differences compared to those trained with GT labels. The largest discrepancy, in the AP_{75} metric, only decreased by 0.5%. It can be seen that the effect of instance segmentation using the labels generated by SAM is equivalent to training the network with the GT. This comparative experiment confirms the usability of pseudo-labels generated by RNP-SAM.

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TABLE VI

RESULTS OF DIFFERENT ORIENTED OBJECT DETECTION METHODS WITH INSTANCE SEGMENTATION ON THE TEST SET OF THE SMALL-DOTA. RED REPRESENTS THE BEST PERFORMANCE. BLUE REPRESENTS THE SECOND-BEST PERFORMANCE

Method	AP	AP_{50}	AP_{75}	AP_{es}	AP_{rs}	AP_{gs}
Rotated MaskRCNN	35.9	70.2	32.9	16.0	36.3	48.0
Rotated Hybrid Task Cascade	38.3	71.8	37.2	16.4	37.9	51.5
Rotated PointRend	34.0	69.2	28.6	15.9	34.7	45.0
Mask roi transformer	38.2	71.0	37.9	17.0	38.0	50.8
Rotated SCNet	34.9	68.3	32.5	12.8	34.8	49.4
RNP-OSOD (proposed)	38.9	72.0	38.7	17.5	39.3	51.7

TABLE VII

RESULTS OF DIFFERENT INSTANCE SEGMENTATION METHODS ON THE TEST SET OF THE SMALL-DOTA. RED REPRESENTS THE BEST PERFORMANCE. BLUE REPRESENTS THE SECOND-BEST PERFORMANCE

Method	AP_{mask}	AP_{mask50}	AP_{mask75}
Mask R-CNN	21.1	42.0	20.5
Mask Scoring R-CNN	21.2	42.3	20.5
Hybrid Task Cascade	22.4	43.8	22.0
YOLACT	12.5	31.0	7.7
PointRend	22.9	45.3	22.4
SOLOv2	8.9	23.4	5.8
SCNet	28.4	50.4	29.5
QueryInst	13.3	33.0	19.0
Mask2Former	14.1	32.3	11.6
CondInst	19.9	42.8	16.8
SparseInst	8.5	24.8	4.0
BoxInst	6.1	19.8	2.5
RNP-OSOD(proposed)	19.2	53.6	22.7

TABLE VIII

EVALUATION OF PSEUDO-MASK EFFECTS UNDER DIFFERENT CONTEXT SIZES ON SMALL-DOTA TRAIN SET. RED REPRESENTS THE HIGHEST RESULT

Enlarge	Context size	mIOU
3×	64	68.84
	128	71.49
	256	73.17
	512	74.75
	1024	73.79
	3×diagonal	71.14
	5×diagonal	74.38

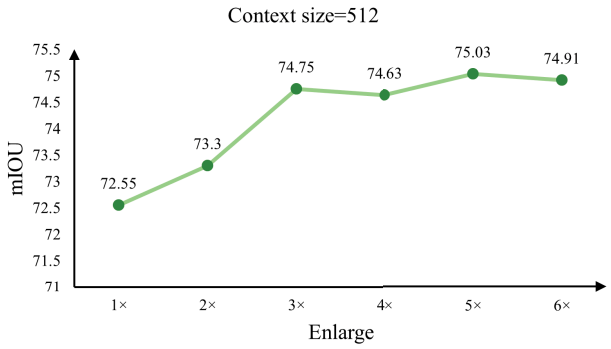


Fig. 12. Evaluation of pseudo-mask effects under different enlargement times on the Small-DOTA train set. The context size is fixed at 512.

3) Comparison With Other Methods: Table V presents a comparison of detection results between the proposed method and several state-of-the-art methods on the Small-DOTA. The comparison methods are consistent with SODA-A. The results show that the proposed RNP-OSOD achieves the highest performance across all evaluation metrics. Additionally, for small-object detection, the two-stage approach outperforms the one-stage method. The proposed RNP-OSOD achieves an AP of 38.9%, surpassing the AP of the highest-accuracy method, RoI Transformer (also the baseline method), by 0.9%. For the three small-object detection metrics: AP_{es} is 0.9% higher and AP_{rs} is 1.0% higher than the highest method, RoI Transformer; AP_{gs} is 0.4% higher than the highest method, ReDet. It can be seen from the table that although the AP improves by only 0.9%, the more stricter evaluation metrics AP_{50} and AP_{75} show an increase of 1.2%. This demonstrates that the proposed RNP-OSOD exhibits robust performance in small-object detection tasks. Table V also shows the detection performance under the SAM2 [70] segmentation model. Both AP and AP_{50} show a 0.2% decrease compared to SAM.

Additionally, Fig. 9 presents a visual comparison of detection results between several typical state-of-the-art methods and the proposed RNP-OSOD method. To provide a more intuitive comparison of detection performance across different approaches, the figure highlights missed detections in yellow and false alarms in red, while magnifying areas where object detection is particularly challenging. Observation reveals that the proposed RNP-OSOD achieves optimal detection performance across diverse scenarios. Among densely packed large vehicles, where instance boundaries are indistinct, other methods exhibit varying degrees of missed detections and false alarms. However, the proposed RNP-OSOD correctly detects these objects, partly due to the supervisory effect of instance segmentation heads, which enhances the network's understanding of object edges. For small objects in complex backgrounds or those with high background similarity and weak responses, the proposed RNP-OSOD achieves detection with fewer errors. The pseudo-masks obtained via RNP-SAM provide auxiliary supervision to the object detection network, effectively amplifying object responses in complex backgrounds while suppressing background interference. This introduces additional enhancement signals for small objects with limited information, significantly improving localization accuracy. Fig. 10 presents the object detection results and corresponding instance segmentation outcomes of the proposed RNP-OSOD. It can be observed that the instance segmentation head, decoupled from the detection head, achieves excellent segmentation performance for both the overall contour and fine-grained structural details of the objects.

Fig. 11 illustrates the precision–recall (PR) curves for each object category in the Small-DOTA dataset, comparing the proposed RNP-OSOD method with several state-of-the-art approaches. The PR curves provide a comprehensive view of the tradeoff between precision and recall for each method across different object classes. It is evident that the proposed RNP-OSOD consistently outperforms other methods across all categories, demonstrating superior detection capabilities. The

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TABLE IX
DETECTION RESULT COMPARISON BETWEEN DIFFERENT ENLARGEMENT TIMES ON SMALL-DOTA TEST SET

Method	enlargement times	AP	AP_{50}	AP_{75}	AP_{es}	AP_{rs}	AP_{gs}
RNP-OSOD (proposed)	3×	38.8	72.1	38.2	17.1	38.9	51.8
RNP-OSOD (proposed)	5×	38.6	71.9	38.5	17.2	38.9	51.4
RNP-OSOD (proposed)	6×	38.8	71.8	38.4	17.3	38.8	51.7

TABLE X
EVALUATION OF PSEUDO-MASK EFFECTS UNDER DIFFERENT SEGMENTATION MODELS ON SMALL-DOTA TRAIN SET

Segmentation Model	mIOU
SAM	74.75
SAM2	72.00

area under the PR curve for RNP-OSOD is notably larger than that of the compared methods, indicating higher overall performance in terms of both precision and recall. This further validates the effectiveness of the proposed approach in accurately detecting small objects in complex remote sensing images.

Similarly, we also conducted comparative experiments on the Small-DOTA dataset using different methods with instance segmentation heads. The results are shown in Table VI. It is evident that the proposed RNP-OSOD significantly outperforms other combined approaches. Its AP exceeds the highest AP among the combined methods (rotated hybrid task cascade) by 0.6%. Additionally, its small-object performance metrics show AP_{es} improved by 1.1%, AP_{rs} by 1.4%, and AP_{gs} by 0.2%.

Moreover, as the proposed method incorporates an additional instance segmentation head, we have also compared the performance of the instance segmentation predictions against current state-of-the-art instance segmentation approaches. These methods encompass not only approaches based on convolutional neural networks, such as Mask RCNN [64], Mask Scoring R-CNN [71], Hybrid Task Cascade [65], YOLACT [72], PointRend [66], SOLOv2 [73], SCNet [67], CondInst [74], SparseInst [75], and BoxInst [76] but also those based on Transformers, such as MaskFormer [77] and QueryInst [78]. To ensure a fair comparison, all methods were trained under pseudo-label supervision generated by RNP-SAM, with metric calculations based on GT masks. The experimental results are presented in Table VII. Although the proposed method exhibits a disadvantage in the instance segmentation metric AP_{mask} compared to some two-stage CNN-based approaches, it demonstrates a clear advantage in terms of AP_{mask50} and AP_{mask75}. Compared to the current best-performing method, AP_{mask50} achieved a 3.2% improvement.

4) Evaluation of Rotation-Normalized Prompting for SAM: The accuracy of the masks generated by RNP-SAM is evaluated by calculating the mIoU with the GT

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N}\text{IoU}_i = \frac{1}{N}\sum_{i=1}^{N}\frac{p_{ii}}{\sum_{j=1}^{N}p_{ij}+\sum_{j=1}^{N}p_{ji}-p_{ii}}. \quad (22)$$

Two critical parameters in RNP-SAM require experimental determination: the object slice size with context and the object slice magnification factor. First, we validated the effectiveness of fixed object slice magnification across different context sizes. To ensure SAM effectively captures object boundaries, providing sufficient surrounding context within the input slice is essential. Based on the 32×32 pixel threshold for small objects, our initial configuration utilized a 64×64 fixed object slice centered on the object to ensure adequate background coverage. We further explored the impact of fixed slice sizes from 128 to 1024. Recognizing the need for flexibility, we introduced a dynamic scaling mechanism where the context range is tied to the object's size. Slice sizes of 3× and 5× the object diagonal length were examined to prevent information loss for extremely small objects. Table VIII shows that when the object slice size is 512, the generated mask achieves the highest mIoU of 74.75 against the GT. Subsequently, to highlight the details of small objects, the object slice must be magnified before rotation. Hence, the effectiveness of fixed object slice sizes at different magnifications was evaluated. The experiment employed six integer magnification factors ranging from 1 to 6. As shown in Fig. 12, magnifying the object slice by 5× yielded the best result at 75.03. Considering that larger slice sizes require more processing time, we further compared detection results under 3×, 5×, and 6× magnification conditions, as shown in Table IX. It can be observed that when the magnification of the section increased from 3× to 5×, AP decreased by 0.2%. When further increased to 6×, AP did not improve. Although some indicators showed slight increases with higher magnification, these improvements were negligible.

At present, SAM2 [70] represents a more advanced segmentation model than SAM. Comparing the performance of these two models when applied to the proposed method constitutes a crucial experiment, facilitating the selection of a superior model for small-object detection in remote sensing. Table X contrasts the segmentation performance of SAM and SAM2 within the RNP-SAM framework under identical parameter configurations. It is evident that SAM achieves an mIoU 2.75% higher than SAM2.

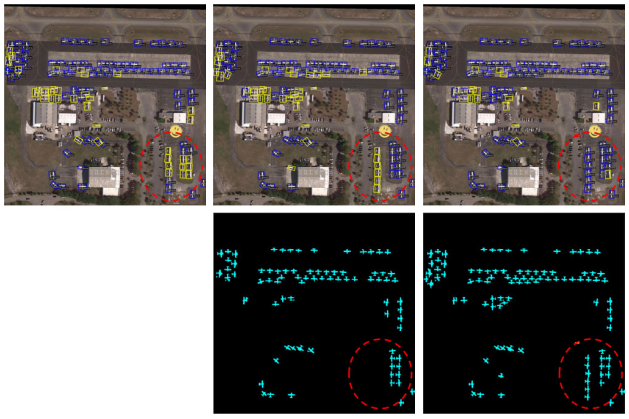
To demonstrate the practicality of RNP-SAM, we tested its processing time after determining the contextual size and magnification of the target section. Here, we observed that RNP-SAM processed 10 000 targets in 41.73 min.

5) Ablation Study for the Proposed RNP-OSOD: Table XI illustrates the effect of incorporating various components of SAIF loss into the proposed RNP-OSOD network on its performance. Fig. 13 visualizes the corresponding detection results and instance segmentation predictions. Table XI indicates that when applying the pseudo-mask generated by

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TABLE XI
INFLUENCE OF DIFFERENT MODULES OF SAIF LOSS ON THE PROPOSED RNP-OSOD

Method	Instance segmentation loss	SAIF loss	AP	AP_{50}	AP_{75}	AP_{es}	AP_{rs}	AP_{gs}	Param.	GFLOPs
			38.0	70.5	37.5	16.6	38.3	50.2	55.06	148.32
baseline	✓		38.2	71.0	37.9	17.0	38.0	50.8	57.84	148.32
	✓	✓	38.9	72.0	38.7	17.5	39.3	51.7	58.35	183.38



(a) baseline (no instance output) (b) baseline + instance segmentation loss (c) baseline + instance segmentation loss + SAS loss (SAIF loss)

Fig. 13. Visualization of ablation study on SAIF loss. (a) Detection result of baseline method (baseline method has no instance segmentation head). (b) Detection and instance segmentation results of baseline with instance segmentation loss. (c) Detection and instance segmentation results of baseline with instance segmentation loss and SAS loss. The red box represents false alarm (FP), and the yellow box represents missing detection (TN). Detection results demonstrate that as the number of supervision modes increases, the omission of small objects in complex backgrounds is significantly reduced. Segmentation results indicate that the model can precisely characterize object shapes and contours.

RNP-SAM solely to the instance segmentation head, AP improves by 0.2%, AP_{50} increases by 0.5%, and AP_{75} rises by 0.4%. For small-object metrics, AP_{es} increases by 0.4%, AP_{rs} improves by 0.4%, and AP_{gs} rises by 0.7%. Overall, network performance improved although the effect was not particularly pronounced. Fig. 13 also shows that while missed detections within the red dashed box decreased, nearly half of the detections were still missed. This indicates that relying solely on auxiliary instance segmentation for network supervision is insufficient to enhance detection for all small objects. While it significantly improves detail detection, in more complex backgrounds, objects resembling the background or those with faint features are largely obscured, resulting in insufficient visibility of details. Applying instance segmentation alone yields limited effectiveness for such objects. However, when applying the SAIF loss, which introduces semantic supervision of features on top of instance segmentation supervision, the result is a 0.9% improvement in AP, 1.5% improvement in AP_{50}, and 1.2% improvement in AP_{75}. As for small object, the result is a 0.9% improvement in AP_{es}, 1.0% improvement in AP_{rs}, and 1.5% improvement in AP_{gs}. In terms of model complexity, incorporating the instance segmentation head increases the model parameters by 2.78M. However, as it does not require output masks during inference, computational complexity remains unchanged. Adding the PCE module further increases the parameters by merely 0.51M, while computational complexity rises by 35.06 GFlops.

TABLE XII
OBJECT DETECTION RESULTS OF USING DIFFERENT MONOTONICALLY DECREASING FUNCTIONS IN SAS

Function	AP	AP_{50}	AP_{75}	AP_{es}	AP_{rs}	AP_{gs}
Linear	38.5	71.2	38.2	16.8	38.4	51.8
Exponential	38.6	71.8	38.1	16.6	38.8	51.5
Power law	38.3	71.0	37.7	16.6	37.9	51.8
Sigmoid	38.9	72.0	38.7	17.5	39.3	51.7

As shown in Fig. 13, only one object was missed within the red dashed area, demonstrating excellent detection performance for small objects in complex backgrounds. This demonstrates that concurrently incorporating supervision signals at both instance and feature levels can more effectively enhance object detection performance. Furthermore, adaptively generating weights within the loss function for objects of varying sizes enables the network to allocate greater attention to extremely small objects. This balances differences among samples of different dimensions, thereby amplifying the response to objects within features. Consequently, the network achieves precise localization of small objects even when confronted with complex backgrounds.

To validate the impact of employing different monotonically decreasing functions in SAS on object detection performance, Table XII presents experiments utilizing four distinct functions. The results demonstrate that the sigmoid function yields the most favorable outcomes, with each metric exceeding the baseline. By utilizing the sigmoid function, a smooth transition within the $(0, 1)$ range is achieved, offering a more nuanced alternative to traditional linear, exponential, or power decay. This mechanism facilitates a clear dichotomy between loss components. It selectively amplifies critical segments while attenuating others, avoiding the pitfalls of indiscriminate decay. Specifically, as losses exceed the object area threshold, the weights asymptotically approach 1 for stable emphasis; conversely, they vanish toward 0 for losses well below the threshold, ensuring rigorous suppression.

To further validate the impact of two hyperparameters in SAIF loss, which are the slope k of the sigmoid function in L_{SAS} and the balanced factor λ of L_{ins} in the loss, the hyperparameter experiments shown in Fig. 14 were conducted. Here, only the AP and AP_{rs} metrics are presented. For the slope k of the sigmoid function in L_{SAS}, three values of 0.01, 0.05, and 0.1 were tested. A larger slope assigns higher loss weights to small objects and relatively lower weights to large objects, better balancing the network's learning for objects of different sizes. It can be observed that the model performs best when k is set to 0.1. For the balanced factor λ of L_{ins} in the loss, five values between 0.5 and 2.5 were tested at 0.5

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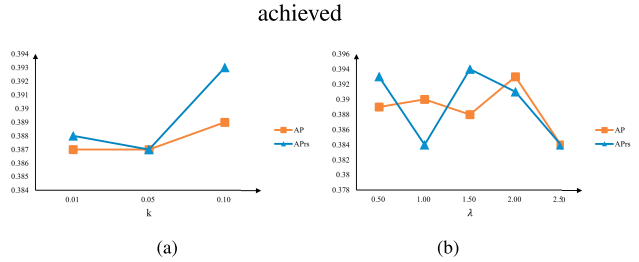


Fig. 14. Hyperparameter experiments of SAIF loss. (a) Influence of slope k of the sigmoid function in L_{SAS} on AP and AP_{rs}. (b) Influence of balanced factor λ of L_{ins} on AP and AP_{rs}.

intervals. The model achieves its highest performance when λ is set to 0.5.

V. Conclusion

In this article, a novel-oriented small-object detection framework called RNP-OSOD is proposed to enhance the discriminative representation of small objects in a complex background. During the first mask generation phase, high-quality object pseudo-segmentation labels are generated by the proposed RNP-SAM. It acquires an object slice based on the center point of the object and the defined context range. These slices are then corrected to a horizontal orientation alongside the rotated bounding box to produce the RNP. During the second mask supervising object detection phase, the pseudo-segmentation labels generated in the first phase are utilized by the proposed SAIF loss to supervise the training of the object detection network from both global and local perspectives. This loss consists of the SAS loss and the instance segmentation loss. The SAS loss handles global feature semantic reconstruction, guiding the model to focus more on small objects by generating loss weights adaptively based on the object areas. The instance segmentation loss is responsible for learning local object detail features. The extensive experiments are conducted on SODA-A [49] and the reconstructed Small-DOTA [68] datasets. Analysis of the generated masks demonstrates that RNP-SAM can produce high-quality pseudo-mask labels and provide pixel-level supervision for the network. The comparison results and detailed ablation results confirm the superiority of the proposed RNP-OSOD framework. In the future, we will explore refined mask generation methods for other annotation formats and investigate more effective mask utilization techniques to enhance detection performance.

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